

Liver Function Tests



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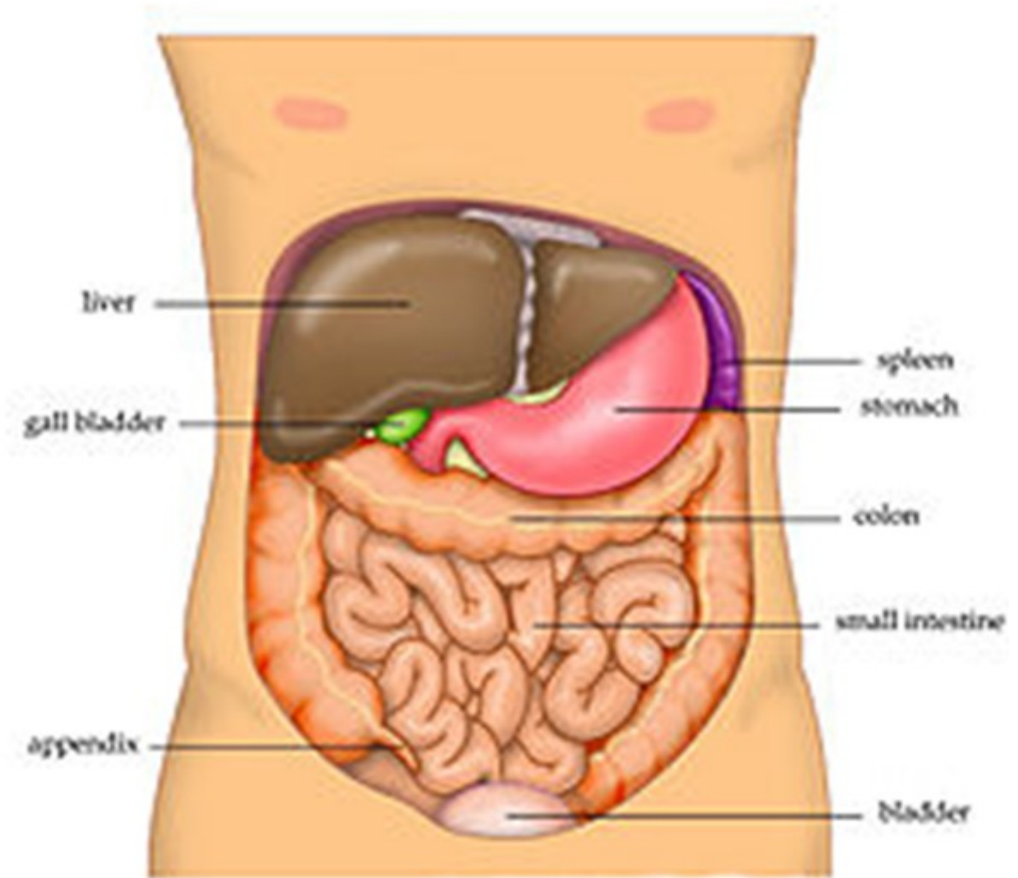
Objectives

At the end of this session, we will be able to:

- Describe the physiological functions of the liver
- Outline the components of liver function test (LFT) and their clinical use.
- Explain the purpose and significance of liver function tests (LFTs).
- Identify key liver enzymes and proteins measured by LFTs.
- Describe how LFT results are interpreted.
- Recognize clinical conditions associated with abnormal LFT results.

Liver

- Liver is the largest and most complex organ of the body
- It is located in the right upper quadrant of the abdomen, below the diaphragm
- Liver has large reserve capacity and capability of regeneration
- It is a multifunctional organ that is involved in diverse body functions



Liver Functions

Synthetic Function: albumin, globulins, cholesterol, triglycerides, lipoproteins, enzymes, hormones, coagulation factors

Metabolic Function: Carbohydrate, protein, and fat metabolism

Detoxification and excretion: urea cycle, bilirubin, cholesterol, drug metabolites

Storage Function: Glycogen, Iron, Vitamins (A, D, E, K and B₁₂)

Production of bile salts: Helps in digestion

Liver Function Tests (LFTs)

What are Liver Function Tests?

- LFTs are a group of blood tests used to evaluate the health and function of the liver.
- They measure the levels of various enzymes, proteins, and other substances in the blood.
- These tests help to detect liver damage, inflammation, and impaired liver function.
- LFTs are a common diagnostic tool in clinical practice, and can also be used for drug safety monitoring.

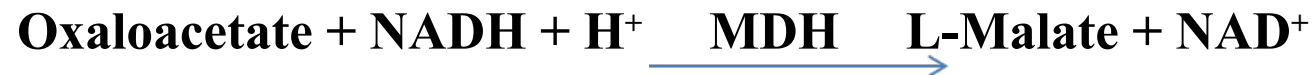
Components of LFTs

Typically, the LFT comprises of:

- Transaminases
 - Aspartate Aminotransferase (AST) &
 - Alanine Aminotransferase (ALT)
- Alkaline Phosphatase
- Gamma Glutamyl Transpeptidase (GGT)
- Bilirubin (Total, Conjugated)

Aspartate Aminotransferase (AST)

- AST (also known as SGOT) is an enzyme involved in amino acid metabolism.
- It catalyses the transfer of an amino group from aspartate to alpha-ketoglutarate.



- Oxaloacetate is reduced to malate-by-malate dehydrogenase (MDH) in the presence of reduced nicotinamide adenine dinucleotide (NADH).
- Decrease in absorbance at 340 nm, due to oxidation of NADH to NAD⁺, is directly proportional to the AST activity in the sample.

CONT...

- AST found in various tissues, including:
 - Liver cells (hepatocytes)
 - Heart muscle
 - Skeletal muscle
 - Kidneys
 - Red blood cells (in lower quantities)
- Thus, AST is not a specific for liver as the ALT.
- It may be elevated in other conditions such as a myocardial infarction and muscle disease
- Measured in blood serum as part of Liver Function Tests.

Clinical Significance

➤ **Normal Range: 8–48 U/L**

➤ **Elevated AST Levels:**

- Indicates cellular damage in organs where AST is found.
- More specifically:

Liver:

- Hepatitis (viral, alcoholic, drug-induced, autoimmune)
- Cirrhosis
- Fatty liver disease (NAFLD/NASH)

- Liver tumors/cancer
- Bile duct obstruction

Cardiac:

- Myocardial infarction (heart attack)
- Myocarditis

Muscle:

- Rhabdomyolysis (severe muscle breakdown)
- Muscular dystrophy

Reduced AST Levels

- Aspartate aminotransferase (AST) levels may decrease due to a number of possible reasons, including:
 - **Vitamin B6 deficiency:** Low AST levels can be a sign of vitamin B6 deficiency.
 - **Kidney disease:** Low AST levels may indicate kidney disease, such as a urinary infection or acute kidney injury.
 - **Liver disease:** Low AST levels may indicate liver disease, such as cirrhosis.
 - **Medications:** Some medications, such as metronidazole (Flagyl) and trifluoperazine (Stelazine), can cause low AST levels.
 - **Uremia:** Uremia, which is when there is urinary waste in the blood, can cause low AST levels

Alanine aminotransaminase(ALT)

- Alanine aminotransaminase (ALAT) is a transaminase enzyme.
- Also called alanine transferase (ALT).

Location of the enzyme

- Cytoplasmic pyridoxal phosphate-dependent (coenzyme).
- ALT is exclusively in cytoplasm of many tissue like liver, heart, kidney, lungs.
 - ❖ But highest concentration is present in liver.

Function

Transamination

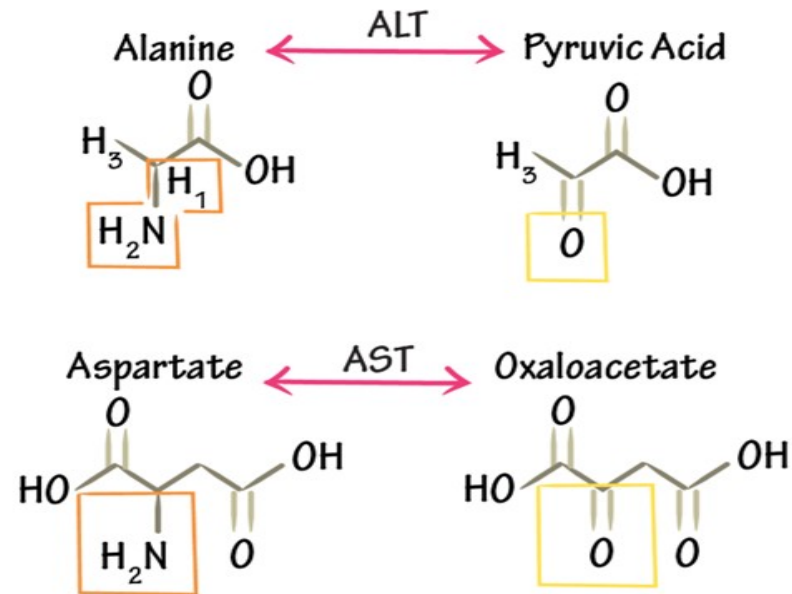
- Transamination in biochemistry is accomplished by enzymes called transaminases or aminotransferases
- It is the reaction between an amino acid and an alpha-keto acid
- The **amino** group is transferred from the former to the latter; this results in the **amino acid** being converted to the corresponding **a-keto acid**, while the reactant **a-keto acid** is converted to the corresponding **amino acid**.

Transamination Components

Aminotransferases

AMINO ACID

α -KETO ACID



- Small amounts of intracellular enzymes are **present in the blood** as a result of **normal cell turnover**.
- When damage to cells occurs, increased amounts of enzymes will be released and their concentrations in the blood will rise.
- However, such increases are not always due to tissue damage.
- Other possible causes include:
 - increased cell turnover
 - cellular proliferation (E.g. neoplasia)
 - increased enzyme synthesis (enzymes induction)
 - obstruction to secretion

Clinical significance

- It is commonly measured clinically as a part of a diagnostic liver function test, to determine liver health.
- Diagnostically, it is almost always measured in units/liter (U/L).

❖ Normal range:

5 u/L to 60 u/L

Uses in liver function test (LFT)

- ALT is one of the sensitive markers that is used as an indicator for Liver disease(along with AST)
- When liver cells are **damaged or destroyed**, the enzymes in the cells **leak out** into the blood, in the blood **raising the enzyme levels** in the blood, where they can be measured by blood test, in which in this case both will be elevated.
- In liver disease the amount of **ALT > AST**.

HEPATOCELLULAR INJURY.

- Some of the reasons the hepatocytes get injured are :
 - Directly from drugs and toxins etc.
 - Hepatic artery damage
 - Hepatic vein damage
 - Biliary duct obstruction

- ❖ When the ALT is elevated, it is important to have an understanding of if it is an **acute enzyme elevation** or **chronic enzyme elevation**.

Acute elevation (> 1000)

- Drug induced liver injury (DILI)
EG : Tylenol (>4g per day)
- Vascular causes
Ischemic hepatitis
- Viral infections (acute hepatitis)
HAV , HEV, HBV
- Autoimmune disease
- Acute biliary obstruction
Cholelithiasis

Chronic elevation (< 500)

➤ Drug induced liver injury (DILI)

EG: Alcohol related ($AST/ALT < 2$)

❖ **The ratio is helpful as long as it is in the lower number (<500).**

➤ Metabolic disease

- Non alcoholic fatty liver disease
- Wilson's disease
- Hemochromatosis.

➤ Viral infection

- HCV

➤ cirrhosis

❖ **Cirrhosis** is a chronic dysfunction of liver over time by causing **excessive fibrosis**, we **lose** cells that have the capability of releasing AST and ALT. drop the levels **will slowly be dropped** over time (pseudo normalization).

Parenchymal injury

➤ Drug induced

- Tylenol (>4g per day), excessive Alcohol use

➤ Metabolic disease

- Alpha 1 antitripsin (AAT) deficiency (decreased level of alpha 1)
- Hemochromatosis (elevated iron levels).
- Wilson's syndrome (elevated copper levels)
- NAFLD (metabolic syndrome)

- Autoimmune hepatitis.
- Biliary obstruction
 - ❖ Bile acids and bile salt will cause a direct injury to hepatocytes.
- Choledocholithiasis
- Ascending cholangitis
- ❖ With biliary obstruction we will get **marked elevation** in alkaline phosphate,
GGT, bilirubin

What does the test result mean?

- Very high levels of ALT (more than 10 times the highest normal level) are usually due to **acute hepatitis, often due to a virus infection.**
- In acute hepatitis, ALT levels usually stay high for about 1-2 months but can take as long as 3- 6 months to return to normal.
- ALT levels are usually **not as high** in chronic hepatitis, often less than 4 times the highest normal level.

❖ Why Get Tested?

- To screen for liver damage and/or to help diagnose liver disease.

❖ Sample Required?

- Serum

❖ When is it ordered?

To evaluate a person who has symptoms of a liver disorder:

- Weakness, fatigue, Loss of appetite
- Nausea, vomiting
- Abdominal swelling and/or pain
- Jaundice
- Dark urine, light colored stool
- Itching (pruritus)

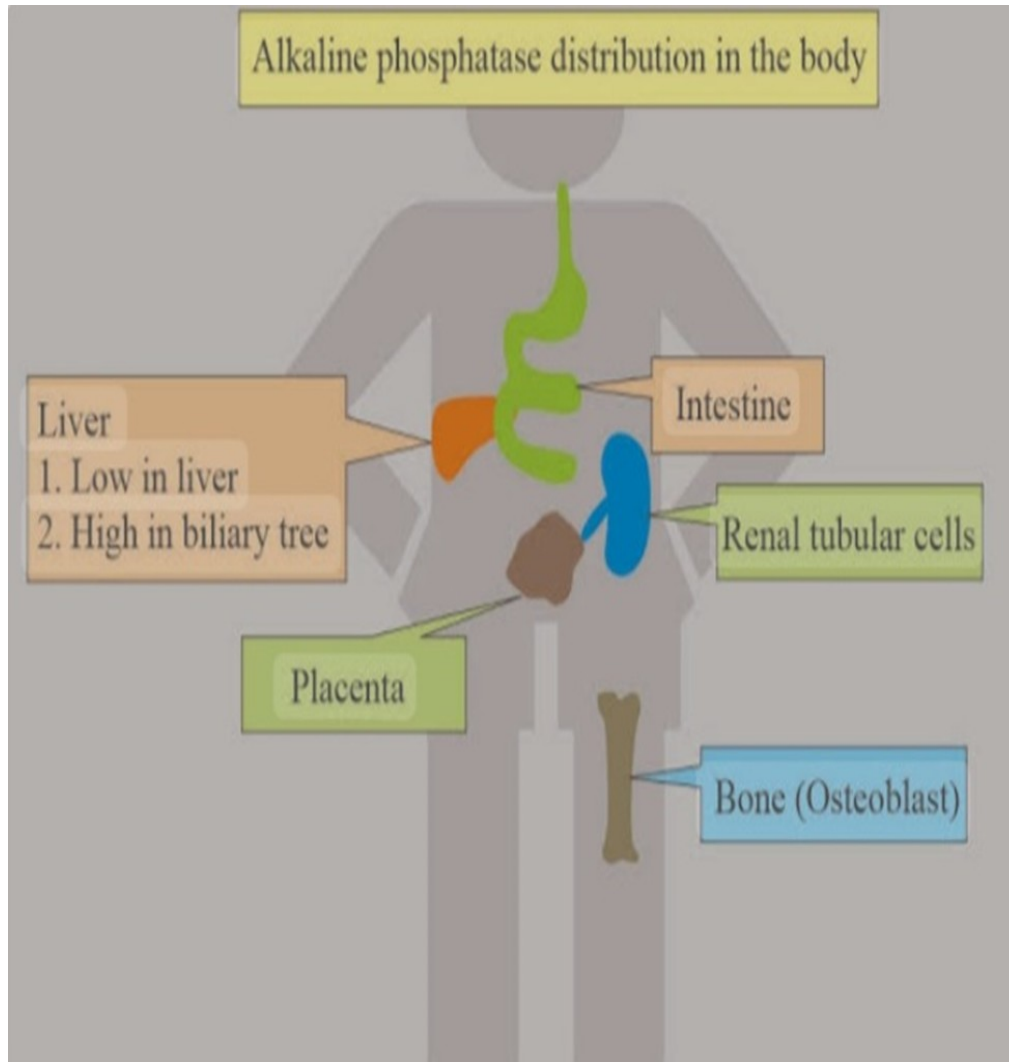
Level of ALT with Cardiac Infarction

- In **severe** myocardial infarction, damage of the liver **due to hypoxia** can increase the ALT level in a few days after onset.

Alkaline phosphatase

- ALP(Alkaline phosphatase)is a hydrolase enzyme responsible for removing phosphate groups from many types of molecule including neucleotides, proteins and alkaloids
- The process of removing the phosphate group is called dephosphorylation.

Location



- ALP present in all tissues but it is particularly concentrated in liver, bile duct, kidney, bones, placenta, intestinal mucosa

What does ALP do in the liver?

- The liver produces ALP to help break down proteins and other organic compounds
- The liver synthesizes more ALP in response to biliary tree obstruction, such as a gallstone or blockage
- The liver's ALP helps transport choline across the bile canalicular membrane

NORMAL VALUES OF ALP

- Normal value for **20-50 years of old man** is 53-128 u/l.
 - Normal values for **20-50 years of old woman** is 42-98 u/l.
 - Normal values slightly different if you **are older than 60** and is in range of 56-119 u/l for man and for woman it is in range of 53-141 u/l.
 - children have high ALP level **because of their growth phase**
- pregnant women typically have higher ALP values due to **contributions from the placenta.**

CONDITIONS IN WHICH IT IS ELEVATED

- Biliary obstruction (blocked bile duct)
- Osteoblastic bone tumor
- Osteomalacia
- Hyperthyroidism
- Hepatitis
- Hyperparathyroidism
- Leukemia
- Myocardial infarction
- Pregnancy
- Osteoblastic bone
- Lymphoma

CONDITIONS IN WHICH IT IS REDUCED

Malnutrition: A lack of protein, calories, or nutrients like zinc, magnesium, vitamin B12/ vitamin C

Hypothyroidism: An underactive thyroid gland

Genetic disorders: Rare conditions like hypophosphatasia, which affects bones and teeth, or Wilson disease, which causes the body to produce too much copper

Medications: Some drugs, including antiresorptives and birth control pills, can lower ALP levels

How is ALP measured?

- An ALP test measures the amount of ALP in your blood
- ALP tests are often part of a routine check up or liver panel test
- You may need to fast for 10 to 12 hours before the test

GAMMA-GLUTAMYL TRANSFERASE

- **Gamma-glutamyl transferase (also γ -glutamyl transferase)** is a transferase that catalyzes the transfer of gamma-glutamyl functional groups from molecules such as glutathione to an acceptor that may be an amino acid, a peptide or water.
- GGT plays a key role in the gamma-glutamyl cycle, a pathway for the synthesis and degradation of glutathione as well as drug and xenobiotic detoxification.



What is Gamma glutamyl transferase test?

- A gamma-glutamyl transferase (GGT) test is a blood test a healthcare provider orders **to check for liver damage or a bile duct blockage.**
- It measures the amount of **GGT.**
- The GGT enzyme is found throughout the body, but it **is mainly found in the liver.**
- If the liver or bile duct is damaged, GGT may leak into the bloodstream. So high levels of GGT in the blood may be **a sign of liver disease or damage to the bile ducts.**

Why is it done?

- **To diagnose liver disease:** Elevated GGT levels can indicate liver damage or disease.
- **To differentiate between liver and bone disease:** While both GGT and ALP may be elevated in diseases that affect the bile ducts and liver, only ALP will be elevated in bone disease.
- ✓ This helps health care providers to understand if the issue may be due to a bone disorder instead of the liver.
- **To monitor alcohol use:** GGT levels can be elevated in people with heavy alcohol use.

NORMAL VALUES OF GGT

- The normal range for adults is **10 to 40 IU/L**. Normal value ranges may vary slightly among different laboratories. Some labs use different measurements or may test different specimens.
- The normal reference range values for GGT is **<30 IU/L** for female.

Conditions in which it is elevated

If the results show higher-than-normal levels of GGT, it may be a sign of liver damage. The damage may be due to :

- ✓ Hepatobiliary obstruction
- ✓ Obstructive jaundice
- ✓ Cirrhosis
- ✓ Alcohol use disorder (AUD)
- ✓ Pancreatitis
- ✓ Diabetes
- ✓ Myocardial infarction
- ✓ Side effects of a medicine.

Conditions in which it is reduce

Alcohol

- Reducing or stopping alcohol consumption can lower GGT levels, even within a month. GGT levels typically return to normal after at least a month of abstinence.

Diet

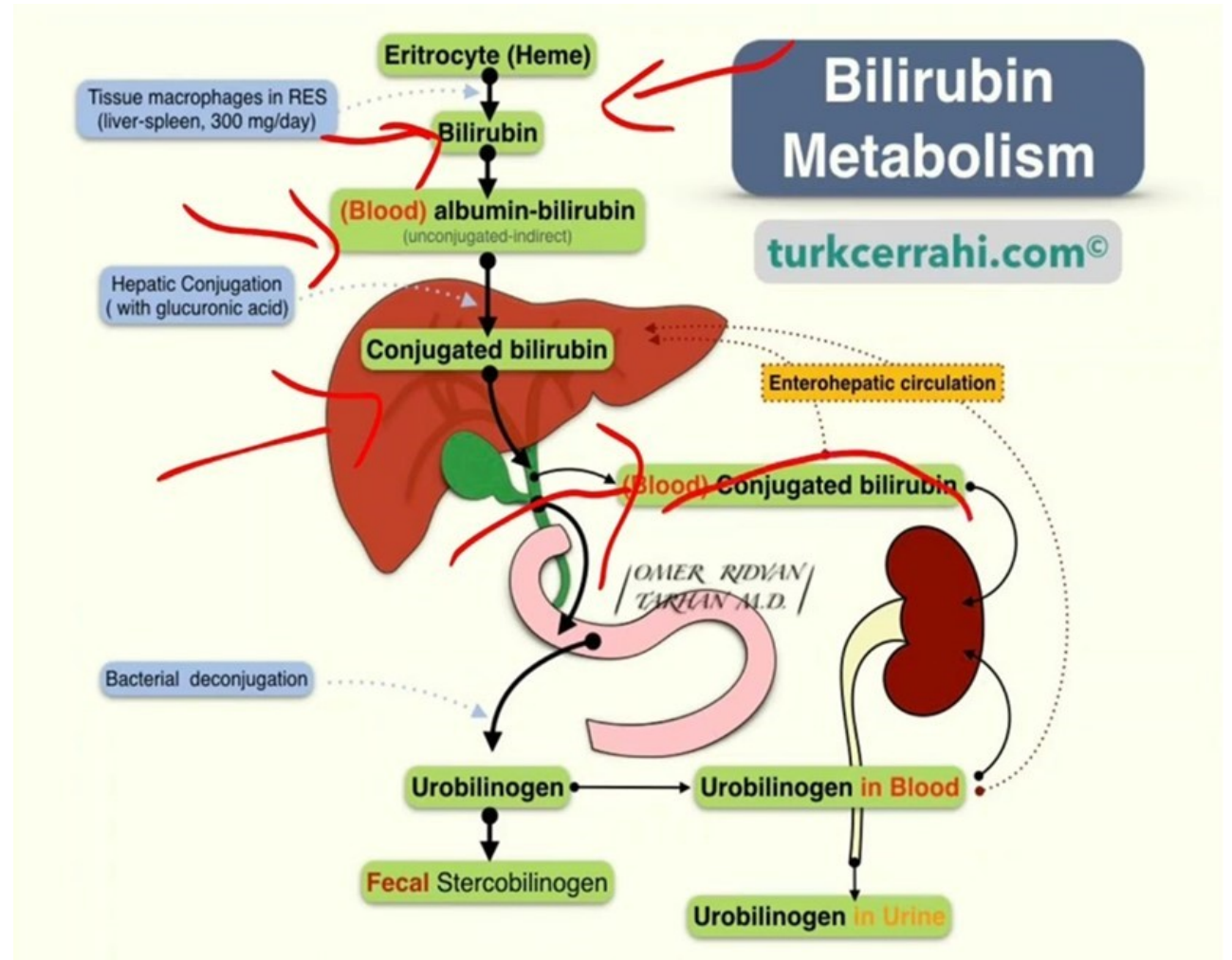
- Eating a diet high in whole foods and low in red meat can help lower GGT levels.

Family conditions

- Some rare conditions, such as familial intrahepatic cholestasis, can cause low GGT levels.

Bilirubin

- Bilirubin is a yellow substance that's produced when red blood cells break down.
- It's a normal part of the body's process of removing waste from old or damaged red blood cells.



How bilirubin is produced and processed?

Hemoglobin, a protein in red blood cells, breaks down.

- ✓ Bilirubin is released from the broken-down red blood cells.
- ✓ The liver receives the bilirubin and releases it into bile and the liver removes most of the bilirubin from the body through bile ducts.
- ✓ Finally, the body excretes bilirubin via bile duct, bilirubin levels is maintained
- ✓

Normal range

A normal range for bilirubin levels in the blood depends on age, gender, and health history.

Adults

- ✓ **Total bilirubin** 0.1- 1.2 (mg/dL)
- ✓ **Direct(conjugated)** bilirubin 0-0.2mg/dl(water soluble)
- ✓ **Indirect bilirubin(unconjugated)** bilirubin 0.2-0.8mg/dl(water insoluble)
- ✓ **Newborns:** Bilirubin levels 1.0 - 12.0 mg/dL.

Liver health and Bilirubin test

Liver health: A bilirubin blood test can help check for liver disease or damage.

Newborns: Bilirubin levels in newborns can be high because they have many red blood cells that are frequently broken down and replaced.

Bilirubin tests - measure the amount of bilirubin in the

- ✓ Blood(plasma and serum)
- ✓ urine

Abnormal findings

High bilirubin levels

- ❖ High bilirubin levels in adults could indicate an underlying health condition, such as jaundice, gallstones, or liver issues.
- ✓ Jaundice- a condition that makes the skin and eyes appear **yellow**.

Low bilirubin levels

- ❖ Certain medications, such as antibiotics, sleeping pills, birth control pills, and seizure medications, can cause lower bilirubin levels.

clinical significance

Total and direct Bilirubin estimation in serum or plasma is used for the diagnosis, differentiation and follow up of jaundice.

A. Hemolytic Jaundice: Increased hemolysis results in elevation of unconjugated Bilirubin level.

B. Obstructive Jaundice: The conjugated Bilirubin increases due to regurgitation of bile into hepatic circulation, because of blockage of bile passage.

C. Hepatic Jaundice: In this case increase of both conjugated and unconjugated Bilirubin serum is estimated in order to assess the extent of liver damage and subsequent progress of regress.

Hyperbilirubinemia

Increases in serum bilirubin (direct + indirect)

Causes:

1)hepatocellular injury

- ✓ inability of liver to conjugate B_l(indirect B_l will be increased)
- ✓ Increase in AST and ALT values

2)Biliary duct obstruction

- ✓ gallstones
- ✓ stricture of bile duct
- ✓ tumor

3)liver disease-viral hepatitis(A,B and c),alcohol, cirrhosis and tumor

4)hemolysis- hemolytic anemia, sickle cell disease

5)physiological neonatal jaundice- due to immature liver function and leading to temporary elevations in BL levels

Treatments

The management of hyperbilirubinemia depends on its cause;

- antiviral medications-viral hepatitis
- surgical intervention-bile duct obstruction
- corticosteroids and transfusions-hemolytic anemia
- phototherapy-neonatal jaundice

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THANK YOU!

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